

Celine Mangahas

714-299-8553 | celinemangahas13@gmail.com | github.com/celinem13 | linkedin.com/in/celine-mangahas

Software Engineer | Python • C++ • Java • Application Development • Linux

Clearance: Previously held Interim Secret

Professional Experience

Software Engineer | Naval Air Warfare Center Weapons Division

Ridgecrest, CA | Jul 2023 – Nov 2023

- Developed Python and SQL workflows to process, standardize, and validate engineering data supporting F-16 simulation and mission-modeling systems.
- Translated manual engineering and data-processing requirements into an automated software workflow, reducing turnaround from weeks to hours.
- Tested, debugged, and validated software and data outputs against known-good results to verify accuracy, consistency, quality, and functionality.
- Collaborated with engineers and technical stakeholders to refine requirements, document software designs and workflows, and support Git-based development and troubleshooting.

Augmented Reality Engineer | Dynamic Augmented Solutions

Rancho Cucamonga, California | Dec 2022 - Feb 2023

- Developed, tested, and debugged C# applications in Unity for Oculus Quest and Microsoft HoloLens, supporting interactive training functionality across multiple hardware platforms.
- Prototyped and tested interactive training features, refining user interactions and visual guidance based on project requirements and team feedback.

Software Engineering Intern | Avenda Health

Culver City, CA | Jan 2022 - Apr 2022

- Developed a user-facing signup experience using JavaScript, HTML, and CSS.
- Supported Python-based healthcare data processing and AI/ML prototyping through dataset validation, debugging, and technical experimentation.

Coding Instructor | Coding Minds Academy

Irvine, CA | Oct 2018 - Sep 2019

- Taught C++, Java, and Python programming concepts including object-oriented programming, data structures, algorithms, debugging, and problem solving.

Technical Projects

Cockpit Telemetry Monitor — C++ / Qt | [GitHub](#) | 2026

- Built a C++ aircraft telemetry-monitoring application using Qt to visualize and monitor simulated telemetry data through a desktop dashboard.
- Implemented telemetry processing, validation, status monitoring, and automated tests using object-oriented C++, CMake, and Linux-based build and debugging workflows.

Expression Parser Using Stacks — Java | [GitHub](#) | 2022, Updated 2026

- Built a Java infix-to-postfix expression parser using generic stack data structures, object-oriented design, operator precedence, associativity, and parentheses handling.
- Improved syntax validation, exception handling, and test coverage for valid expressions, malformed input, and edge cases.

NerdNest — Full-Stack Web Application | [GitHub](#) | 2026–Present

- Developing a full-stack application using React, JavaScript, Node.js, Express, MongoDB, and REST APIs with JWT authentication and user profile management.
- Building frontend/backend integrations, validation, error handling, application state, data models, and database persistence.

Skills

Languages

Python • C++ • Java • SQL • C# • JavaScript • C

Application Development

Qt • CMake • Node.js • Express • REST APIs • JSON

Databases & Data

SQL Server • MySQL • MongoDB

Software Engineering

OOP • Data Structures & Algorithms • Debugging • Testing • Requirements Analysis

Tools & Practices

Git • GitHub • Linux • Jira • Agile/Scrum • Code Reviews

Education

California State Polytechnic University, Pomona | Bachelor of Science in Computer Science | 2018–2022